

- 2) Write a Python function that takes a number as a parameter and check the number is prime or not.
- 3) Write a Python function to calculate the factorial of a number (a non-negative integer). The function accepts the number as an argument.
- 4) Write a Python function that accepts a string and calculate the number of upper case letters and lower case letters.
- 5) What is the output of the following program?

```
num = 1
def func():
    num = 3
    print(num)
func()
print(num)
```

[Space for Answers]

1) Hello
Hello
Hello
Hello

2) Refer to attachment

3) Refer to attachment

4) Refer to attachment

5) 3
1

problem: Function Positional / Required argument, Function with Keyword argument, Function with default argument, Function with variable

(X) Practical related questions (Teacher may assign more questions related to practical. Students may add extra pages for answer if needed)

- 1) What are positional arguments?
- 2) Write program to demonstrate use of variable length arguments.
- 3) What is a Function Argument in Python?
- 4) Explain function with keyword argument.
- 5) What will be the output of following code?

```
def print_animals(*animals):
    for animal in animals:
        print(animal)
print_animals("Lion", "Elephant", "Wolf", "Gorilla")
```
- 6) What will be the output of following code

```
def print_food(**foods):
    for food in foods.items():
        print(food)
print_food(Lion="Carnivore", Elephant="Herbivore", Gorilla="Omnivore")
```

[Space for Answers]

1) Positional arguments are arguments passed to a function in the same order as the parameters defined in the function. The value of each argument is assigned based on its position.

```
2) def add-numbers(*nums):
    total = 0
    for i in nums:
        total = total + i
    print("Sum = ", total)
add-numbers(10, 20)
add-numbers(5, 10, 15, 20)
```

3) Function arguments are the values supplied to a function during a function call to provide input to the function.

4) Keyword arguments are arguments passed to a function using the parameter name and value, allowing the arguments to be given in any order.

5) Lion
Elephant
Wolf
Gorilla

6) ('Lion', 'Carnivore')
('Elephant', 'Herbivore')
('Gorilla', 'Omnivore')

(XI) References/Suggestions for further reading

1. <https://www.geeksforgeeks.org/keyword-and-positional-argument-in-python/>
2. <https://stackoverflow.com/questions/5940180/python-default-keyword-arguments-after-variable-length-positional-arguments>
3. <https://www.geeksforgeeks.org/variable-length-argument-in-python/>
4. <https://www.toppr.com/guides/python-guide/references/methods-and-functions/function-argument/python-function-arguments-default-keyword-arbitrary/>
5. <https://www.youtube.com/watch?v=OCbSteqZsKQ>

(XII) Assessment Scheme (50 Marks)

S. No.	Weightage- Process related: 60%	Marks-30
1.	Logic formation: 40%	
2.	Debugging ability: 10%	
3.	Correctness of program code: 10%	
	Weightage- Product related: 40%	Marks-20
4.	Expected output: 15%	
5.	Timely completion of practical: 15%	
6.	Answer to sample questions: 10%	
	Total 50	
	Dated Signature of Course Teacher	

Observations : _____


```
def even_check(num):  
    if num % 2 == 0:  
        return True  
    else:  
        return False  
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  
even_numbers = filter(even_check, numbers)  
print(list(even_numbers))
```

[Space for Answers]

- a) The filter() function in Python is used to select elements from an iterable (like list or tuple) that satisfy a given condition.
- b) The map() function in Python is used to apply a given function to each element of an iterable (like a list or tuple) and return the results.
- c) 120
- d) [1, 4, 9, 16, 25]
- e) [2, 4, 6, 8, 10]